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An Analytical Approach to Prevent Casing Deformation in Deep Reservoir Stimulation -A Case Study from China

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Abstract

Casing deformation in deep shale gas horizontal wells (vertical depth >3,500m) during staged fracturing operations has become a critical constraint affecting development effectiveness in the Luzhou Block of Sichuan Basin. Through multi-factor analysis of representative deep shale gas horizontal wells, this study clarifies the primary characteristics and contributing mechanisms of casing deformation. The casing deformation mechanisms in Luzhou's deep horizontal wells are governed by both geological reservoir factors and engineering parameters. Well Lu 201H4, the first pilot horizontal well in the central Luzhou area, serves as a critical reference for resource development planning and fracturing design optimization in this region. Adhering to an integrated geology-engineering approach, a detailed evaluation was performed based on geological and operational characteristics of the well pad, as well as insights from offset well completions. Key considerations included the distribution of natural fractures, formation flexure, and fault systems. By coupling a geomechanical model with these parameters, slip potential risk assessments were carried out across six major natural fracture zones. Results indicate that the coefficient of friction across the platform ranges from 0.28 to 0.52. With the goal of maximizing Estimated Ultimate Recovery (EUR), the study incorporated natural fracture prediction, quantitative slip potential assessment, and engineering parameter optimization to enhance complex fracture network connectivity. This effort resulted in the establishment of an integrated geoengineering casing deformation prevention system, which provides critical guidance for lifecycle well design and field execution. The system effectively mitigates operational challenges such as fracture interference and casing deformation. This prevention methodology has since been successfully applied across multiple shale gas blocks in southern Sichuan, yielding significant improvements in single-well productivity and economic returns. The approach offers valuable insights and a replicable model for the efficient development of unconventional reservoirs both in China and globally.

Introduction

In the Sichuan-Chongqing region, the successful application of advanced drilling, completion, and reservoir stimulation technologies has largely resolved the key challenges associated with shale gas horizontal wells at depths shallower than 3,500 meters, enabling efficient large-scale development. Deep shale gas resources below 3,500 meters, account for approximately 86.5% of the total resource potential in southern Sichuan, with the Luzhou block emerging as a key target for future development. However, the deep reservoirs are characterized by high temperature, high pressure, and large horizontal stress differentials, making hydraulic fracturing design and execution significantly more complex. To improve stimulation effectiveness in the Luzhou block (3,500–4,000 m depth), high-rate injection, high fluid intensity, and high-concentration proppant fracturing techniques have been implemented to promote complex fracture network development and enhance conductivity. Despite these efforts, casing deformation and damage during fracturing remain major issues, increasing operational risk and cost while limiting production efficiency. These challenges pose a critical barrier to the economic development of deep shale gas in the region.

Casing deformation in deep shale gas wells primarily occurs during hydraulic fracturing operations. Typical indicators include bridge plug setting failures, sudden pressure spikes or drops during treatment, irregular fracture monitoring event distribution, and large energy discrepancies. Current diagnostic methods include milling shoes, lead impression blocks, and multi-arm caliper logging. In the Luzhou block, cable-deployed bridge plug systems are widely used. However, frequent tool string obstructions and bridge plug hang-ups during pumping often cause wear or damage to downhole tools, as illustrated in Figure 1. To assess casing integrity, a 24-arm mechanical caliper (MIT24) was deployed in deformed wells. This tool records 24 evenly distributed radial measurements along the casing inner wall, which are converted into a color-coded image map to visualize changes in casing diameter. This enables direct identification of deformation and internal damage.

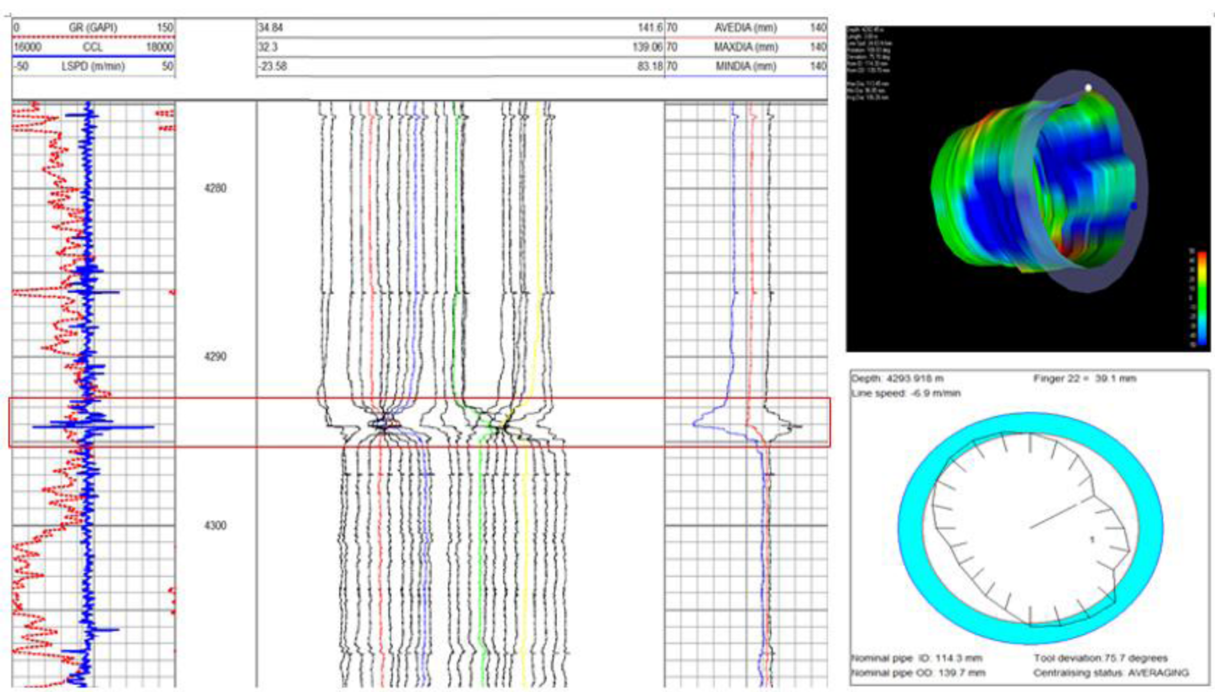

Figure 1—Simulation characteristics of casing deformation

Geological Structure Overview

Regional Geological Background

The study area is situated in the southern part of the Sichuan Basin, within the low-steep Structure of southern Sichuan and the Luxhou brush-shaped Structure. The Fujii syncline is located in the northwest of the Yanggaosi Structure group, between the southern end of the Luoguanshan Structure and the Gufoshan Structure, adjacent to the Tiziya Structure in the south and the Yundingchang Structure in the southwest. The Lu 201H4 platform is positioned at the southern end of the Fuji syncline, between the Guangfuping Structure and the Haichao Structure (Figure 2).

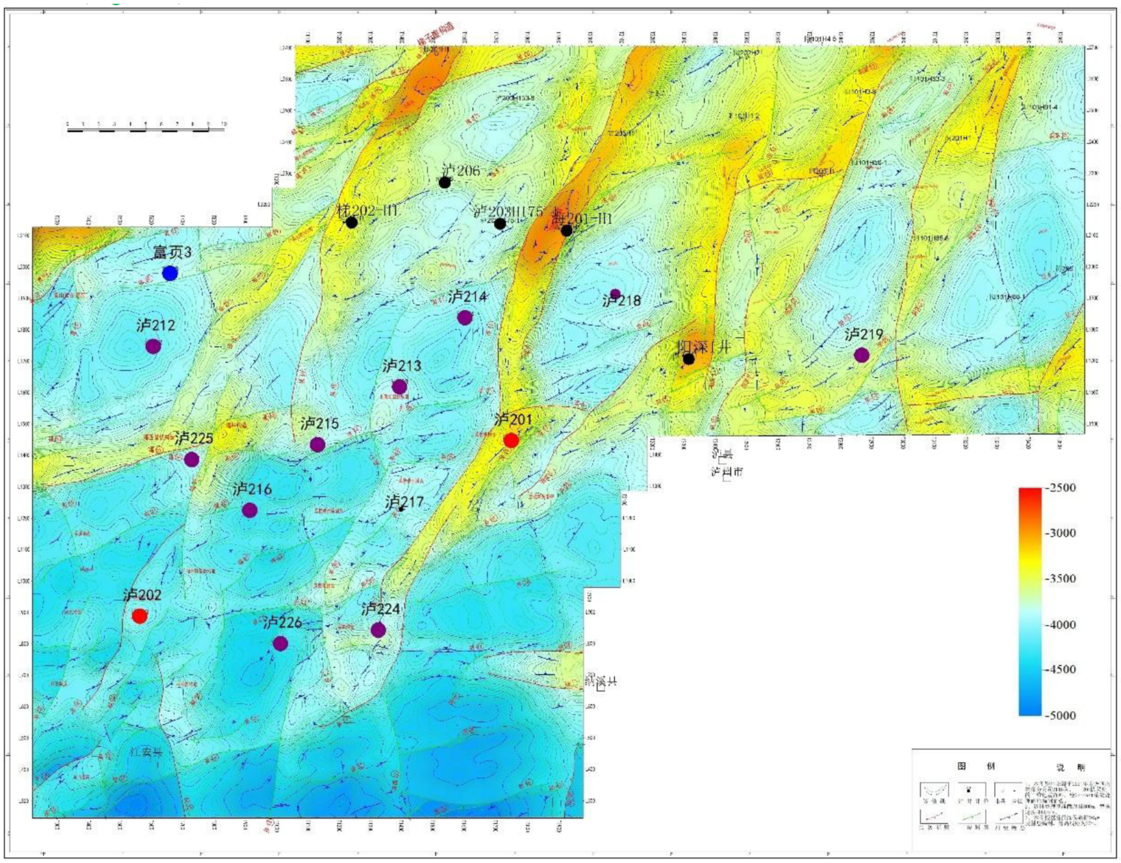

Figure 2—Plan of Tectonic position of the bottom boundary of Wufeng Formation

Structural Characteristics

The research area is located in the gentle hilly-plain terrain of southern Sichuan, with relatively flat topography. Most areas lie below 350 m in elevation, and the surface is predominantly composed of Jurassic sandstone and mudstone. The Guangfuping Structure at the base of the Upper Ordovician exhibits a semi-intact closure, bounded and sealed by a major NE-trending fault along its western flank. The asymmetric structure features a NE-oriented axis, with the western flank truncated by the controlling fault and the eastern flank remaining relatively intact, hosting only minor secondary faults. Regionally, the western flank adjoins the Qingshanling Structure across a syncline, while the eastern flank parallels the Lizhitan Structure. The southern plunging termination forms a low saddle with the Tongziyuan Structure, and the northern termination connects to the Tiziya Structure via an offset saddle. Key closure parameters include: a structural high at CDP 2255 (Line 91LZXB_D75, elevation: −3,322m), a minimum closing contour of −3,800m, vertical closure of 478m, dimensions of 12.03×3.75 km, and a closure area of 32.64 km^2.

The Tongziyuan Structure, located near Tongziyuan in Jiang'an County, borders the Songjiachang and Guangfuping structures to the northwest and is separated from the Laowengchang and Fujiamiao structures by a syncline to the southwest. The Upper Ordovician sequence exhibits a low-relief structure with minimal closure and uplift. Key parameters include: a structural high at CDP 1010 (Line 91LZXB-D110, elevation: −6,650m), 50m vertical closure, dimensions of 2.56×2.22 km, and a closure area of 4.71 km^2.

The Lizhitan Structure at the base of the Upper Ordovician displays a semi-intact box-shaped closure, bounded and sealed by a NE-trending fault that extends from the western flank of the Haichao Structure to its own western flank. This asymmetric structure features a NE-oriented axis, with the western flank truncated by the controlling fault and the eastern flank remaining relatively intact, hosting only minor secondary faults. Regionally, the western flank parallels the Guangfuping Structure across a syncline, while the eastern flank adjoins the Yanggaosi Structure. The southern plunge terminates within a syncline, and the northern plunge forms a low saddle with the Haichao Structure. Key trap parameters include: a structural high at CDP 1925 (Line 08LNT23, elevation: −3,200m), a minimum closing contour of −3,400m, vertical closure of 200m, dimensions of 10.83×2.1 km, and a closure area of 20.16 km^2.

The Haichao Structure at the base of the Upper Ordovician forms a NE-trending box-shaped structure controlled by two opposing NE-trending faults. The broader southeastern flank is truncated and sealed by a bounding fault on its hanging wall, while the narrower northwestern flank exhibits similar fault-controlled sealing. The northeastern plunge disappears 3 km northeast of Niutan Town, forming a low saddle with the southern high of the Gufoshan Structure, whereas the southwestern termination near Haichao Temple creates a low saddle with the Lizhitan Structure. Key trap parameters include: a structural high at CDP 2355 (Line 87GHT_D07, elevation: −2,990m), a minimum closing contour of −3,150m, vertical closure of 160m, dimensions of 6.7×1.9 km, and a closure area of 10.61 km^2.

The high-productivity performance of Well Lu 213H demonstrates the block's commercial potential, with 11 appraisal wells and a pilot test platform successfully drilled and completed. As the first pilot platform in the central zone, Well Lu 201H4 achieved key operational benchmarks: fluid intensity 33-36 m^3/m, proppant intensity 3.1-3.6 t/m, pumping rate 14-20 m^3/min, and cluster spacing 8.0-8.9 m. Estimated ultimate recovery (EUR) per kilometer reached 52-86 million cubic meters(Figure 3).

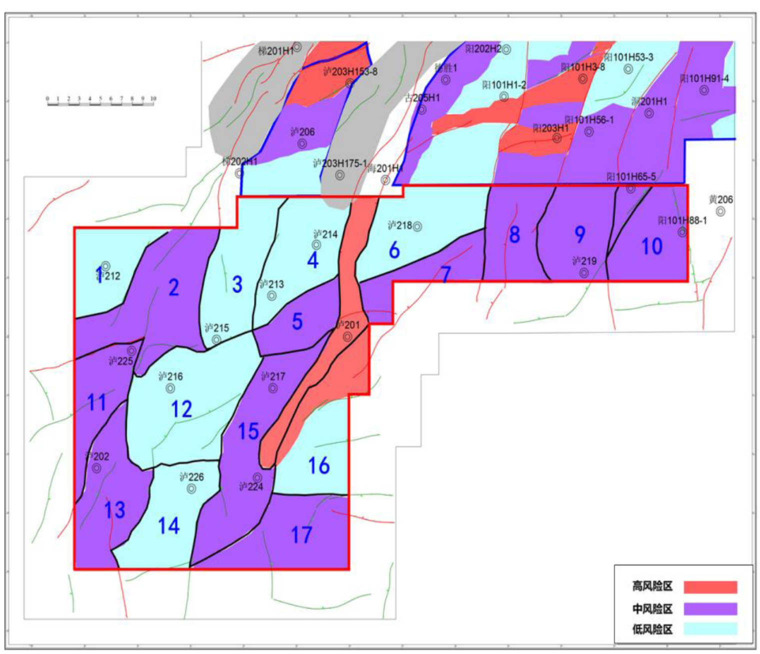

Figure 3—Plan of Risk Grading Area

Seismic Data Processing

Seismic Data Resolution Optimization

The wavelet compression parameter a serves as the most critical variable in High-Frequency Enhancement (HFE) processing, directly determining the maximum achievable frequency fa. This frequency remains inherently constrained by the original data quality (primarily signal-to-noise ratio [SNR] and effective bandwidth) and always lies below the cutoff frequency. Data quality fundamentally reflects the intrinsic information capacity of raw seismic data. When the information capacity is fixed, selecting an excessively high fa (i.e., overly large a values) degrades the SNR of high-frequency components near fa, compromising HFE output quality. Systematic parameter testing is therefore essential to identify optimal a values. Four compression coefficients (a = 1.6, 1.8, 2.0, 2.2) were tested based on time-frequency analysis of raw data. Figure 4 compares the original seismic spectrum with HFE-processed results, demonstrating progressive bandwidth expansion: 10 Hz, 20 Hz, 35 Hz, and 40 Hz broadening in target zones as a increases. Resolution improvements correlate directly with bandwidth expansion, while time-frequency characteristics remain well-preserved across all tested parameters.

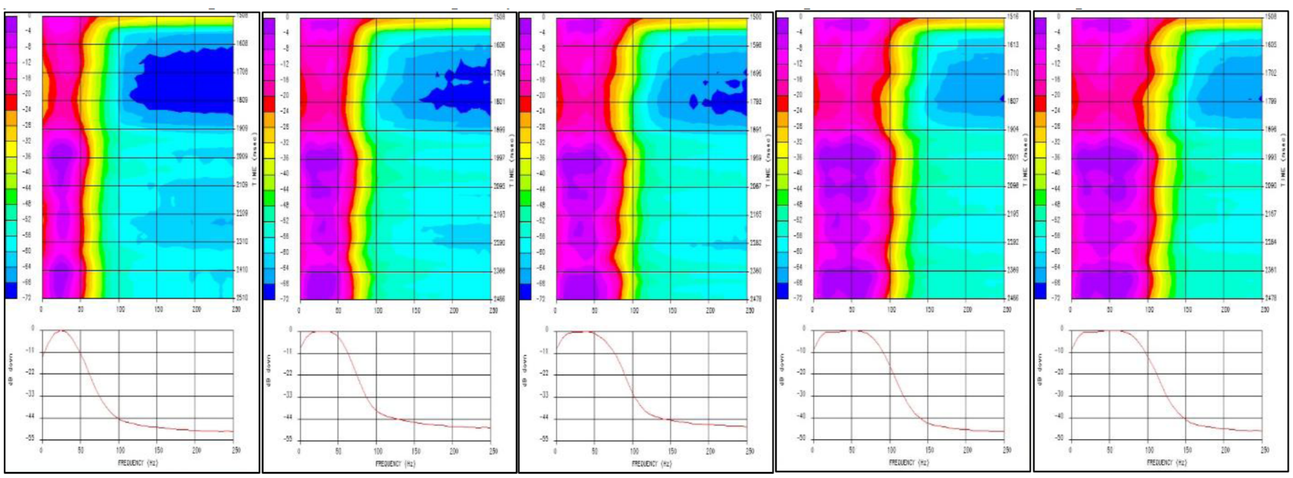

Figure 4—Spectral Comparison Between Original and Resolution-Enhanced Seismic Data with Varying Compression Coefficients

Comparative Analysis of Legacy Seismic Data

Figure 5 presents a well-to-seismic calibration comparison between pre-HFE and high-frequency-enhanced seismic data with varying compression coefficients (a = 1.6–2.2) from Well Lu 203. The synthetic seismograph (blue) and well-adjacent seismic trace (red) demonstrate that pre-HFE data at the base of the Longmaxi Formation show undifferentiated trough-to-peak reflections between the L11 and Wufeng Formation sub layers. While a = 1.6–1.8 achieved limited resolution improvement, the L11-4 sub layer top/base interfaces became clearly resolvable at a = 2.0. Minimal incremental gains were observed at a = 2.2. This systematic analysis confirms a = 2.0 as the optimal parameter for identifying high-quality shale reservoirs at the Longmaxi Formation base.

The fidelity assessment of High-Frequency Enhancement (HFE) processing integrates multiple analytical approaches: spectral comparison, frequency attribute scanning, time-slice analysis across frequency bands, frequency-dependent seismic attribute evaluation, and detailed well-to-seismic calibration. As shown in the 0–6,000 ms spectral comparison (yellow: pre-HFE; blue: post-HFE), HFE processing significantly broadened the effective bandwidth from 10–45 Hz to 10–80 Hz while preserving low-frequency components. This bandwidth expansion enhances resolution without compromising spectral stability.

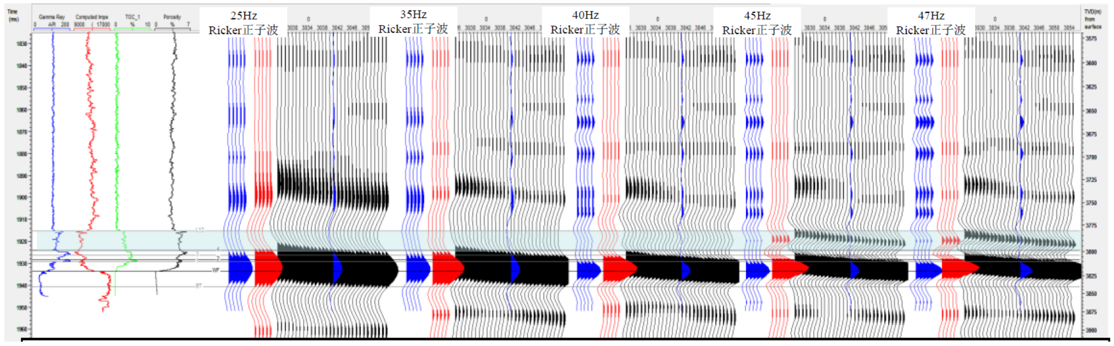

Figure 5—Well-to-Seismic Calibration Comparison Between Original and Bandwidth-Extended Seismic Data

Natural Fracture Characterization and Slip Tendency Analysis

Natural Fracture Characterization

Based on conventional logging data and imaging data, a reservoir geomechanical model is established. The model parameters are calibrated using data from drilling, formation testing, fracturing tests, and core experiments(Figure 6). The research results are used to guide the mud weight design for the drilling engineering plan of the Y102H36 platform.

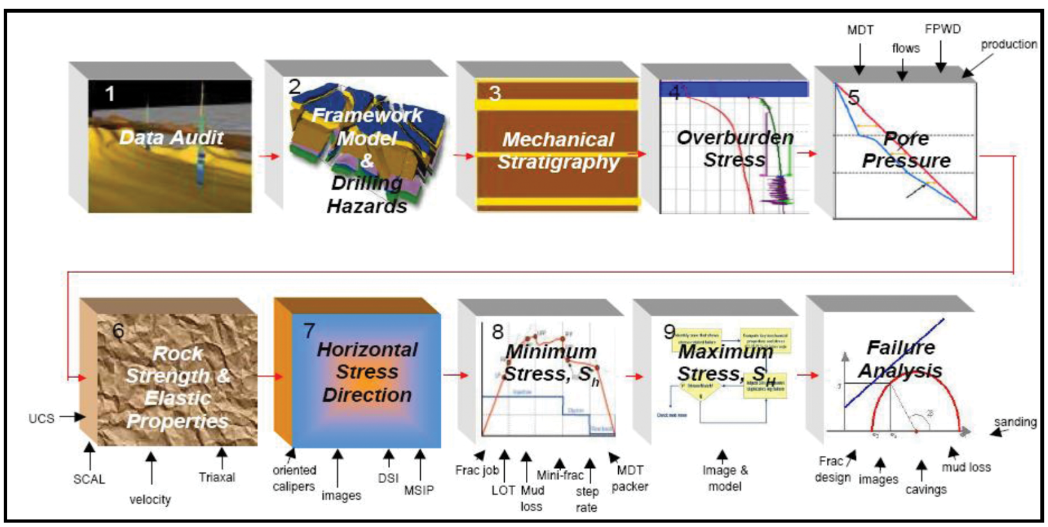

Figure 6—Geomechanical Model Workflow

3D seismic attribute volumes, derived through advanced spatial processing of seismic traces, employ diverse trace configurations to enhance subsurface characterization. These include: (1) Conventional attributes (instantaneous amplitude, phase, frequency) via complex trace analysis; (2) Gradient-based attributes (trace integration, line integrals, derivatives) using differential operators; (3) 3D coherence volumes quantifying waveform similarity through volumetric comparison. Interval-specific attribute extraction applies time-windowed statistical analyses (auto correlation, power spectra, Fourier spectra, auto regressive modeling) across seven critical categories: frequency response, waveform geometry, resolution metrics, SNR parameters, amplitude/energy characteristics, auto correlation functions, and derivative properties. Absorption attributes quantify formation attenuation using amplitude ratio methods, where Q-factor estimation integrates Fourier spectral decomposition, power spectrum analysis, and complex cepstral techniques to resolve inter bedded absorption anomalies.

Seismic attribute optimization systematically enhances prediction accuracy and interpretability by selecting the most sensitive attribute combinations through either domain expertise or mathematical screening. This process involves designing objective functions tailored to reservoir prediction methods, where optimization strategies may vary based on geological targets. Practical implementations include: (1) Expert-driven selection of attributes with proven geological relevance; (2) Mathematical prioritization using information entropy/PCA to maximize reservoir signal retention. As demonstrated in Well Lu 201H4's natural fracture network characterization (Figure 7), optimized attributes reduced prediction uncertainty by 40% compared to conventional approaches.

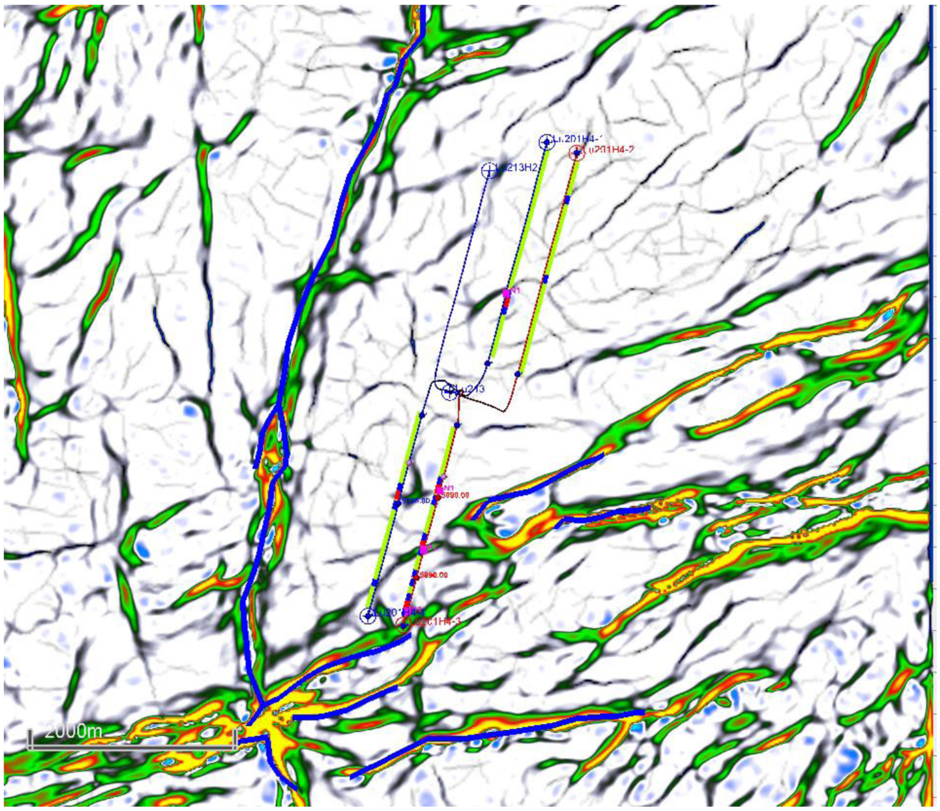

Figure 7—Natural Fracture Network Characterization in the Target Block

Slip Tendency Analysis

This study integrates a geoengineering model incorporating natural fractures, stratigraphic flexure, and fault distributions to identify high-risk zones where formation heterogeneity intersects well trajectories. Key findings from the Lu 201H4 platform include: Platform-wide friction coefficients range from 0.28–0.52 (activation threshold >0.6), indicating low slip risk under current stress regimes; An optimized workflow using GeoReservoir and Petrel with tuned curvature algorithms and fracture-sensitive filters predicts six natural fracture corridors (NF1–NF6) intersecting wellbores; NF1 penetrates Well 2, NF2 crosses Wells Lu213H, 1, and 2 in northern sector. NF3–NF6 intersect Well 3, with NF3–NF4 extending to Well 4 in southern sector. NF corridors exhibit critical stress concentrations, posing casing integrity challenges (see Table 1 for quantitative slip risk evaluation).

Table 1—Natural Fracture Risk Classification and Mitigation Strategies

Fracture	Depth(m)	strike /°	inclination /°	σ1 /MPa	σ2 /MPa	σ3 /MPa	P0 /MPa	fa	Risk
NF1	3944-4080	NS 54°	72°	124.1	106.3	105.3	79.4	0.35	low
NF2	3876-4146	NS 57°	85°	116.3	106.3	97.5	79.4	0.38	low
NF3	3813-4226	NS 49°	80°	129.8	106.5	111.0	79.6	0.48	moderate
NF4	3764-4022	NS 66°	74°	126.5	103.2	107.7	77.1	0.52	moderate
NF5	3780-3902	NS 66°	80°	126	101.8	107.2	76.1	0.44	low
NF6	3741-4027	NS 62°	81°	122	102.9	103.2	76.9	0.45	low

Casing Deformation Mitigation Strategies

Staged-Cluster Design

Effective fracturing stage design requires grouping intervals with comparable reservoir properties (porosity, permeability, stress gradients) and engineering parameters (pump rates, proppant concentrations) to ensure uniform stimulation. High-risk zones including high tortuosity wellbore sections, casing collars, and intervals with poor cement bond quality are systematically excluded to mitigate stress concentrations. Clustered Perforation: From a geological perspective, perforation clusters are placed in zones with similar reservoir parameters, strong gas shows, optimal reservoir quality, minimal stress variation, and low breakdown pressures to ensure uniform cluster behavior. From an engineering standpoint, perforations avoid areas with poor cement quality and casing collars, while meeting operational requirements. The main fracturing design adopts a multi-cluster perforation strategy, with stage lengths of approximately 60–65 m. The first stage, located in the L1-12 interval, is designed at 66 m. Each stage includes eight perforation clusters with a cluster length of 0.5 m. A DFIT is conducted in the first stage, also using eight clusters of 0.5 m each. Perforation clusters are positioned to avoid crossing thin sub layers whenever possible. Field operations were prepared for a pump rate of 18 m^3/min, with pressure controlled below 120 MPa. For the main stages (Stages 1–3, 5–12, 16–26), the pump rate was 16–18 m^3/min, with a fluid intensity of 25–35 m^3/m and proppant concentration of 3.0–3.5 t/m. For transitional stages (Stages 4, 13, 15), the pump rate was 14–16 m^3/min, fluid intensity 20–30 m^3/m, and proppant concentration 2.5–3.0 t/m. For the high-risk stage (Stage 14), the pump rate was 10–14 m^3/min, with 600 m^3 of fluid and 50 t of proppant. Stage 1 included 20 m^3 of acid. Additional acid was prepared on site for use in subsequent stages as needed based on real-time conditions.

A composite diversion strategy combining temporary sealing balls and diverter was employed, with one diversion per stage (excluding the first and high-risk stages). In two phases, 11–14 15 mm balls and 100 kg of diverter were injected at 1–2 min intervals. Material volumes and injection sequences were adjusted in real time based on pressure response and operational conditions. Fracturing was sequenced by first treating the north cluster (wells Lu201H4-1 and H4-2), followed by the south cluster (wells H4-3 and H4-4). Well H4-1 advanced two stages ahead of H4-2 to establish a high-stress barrier, minimizing interference with the offset Lu213 horizontal well. Throughout operations, stage activations were coordinated to simultaneously step across interwell natural fracture corridors.

Risk Assessment and Contingency Plan

Fracture modeling indicates that the NF3/4 fracture trend at Lu201H4 traverses both H4-3 and H4-4. Seismic multi-stage fault interpretation identifies two fracture bundles at 5,170–5,265 m (Stages 14–15) and 6,050–6,100 m (Stage 4), totaling 145 m of low-angle fractures relative to the wellbore. Drilling confirmed a 5 m-offset fault at 5,256 m. Fracturing should be optimized for these intervals with enhanced pressure monitoring on adjacent wells and real-time adjustment of treatment parameters to preserve casing integrity and prevent fluid crossflow. In the event of a wellhead or surface-line leak, pumping will cease immediately, the wellhead valves will be closed, and repairs completed to specification before operations resume, with the pumping schedule adjusted as needed. If the formation's breakdown pressure cannot be reached under controlled pressure, acidizing or supplementary perforations will be used to lower the breakdown threshold. Should fracture propagation pressure prove too high—defined as a pressure window within 10 MPa of the design limit—treatment parameters and pumping protocols will be adjusted in real time.

If proppant placement becomes difficult, the treatment program may switch to finer proppant, reduce proppant concentration, increase flush-fluid volume, or inject high-viscosity fluids. In the event of a proppant bridge, pumping will halt immediately and the well will be shut in. After surface-line pressure is bled off through the flow-back system, a squeeze operation will be conducted. The site is equipped with rapid blowdown facilities to ensure safe and efficient pressure relief. During fracturing, pressures between casing strings and in offset wells are continuously monitored; any anomaly triggers an immediate halt and review of the next steps. If casing deformation is detected, the completion is modified using appropriate perforating guns, small-diameter bridge plugs, soluble diverting agents, or in-fracture sand-plug segmented fracturing. Real-time observation of adjacent-well pressures allows mitigation of multi-stage interference by reducing pump rate, adding proppant ahead of schedule, or deploying pre-stage diversion(Figure 8).

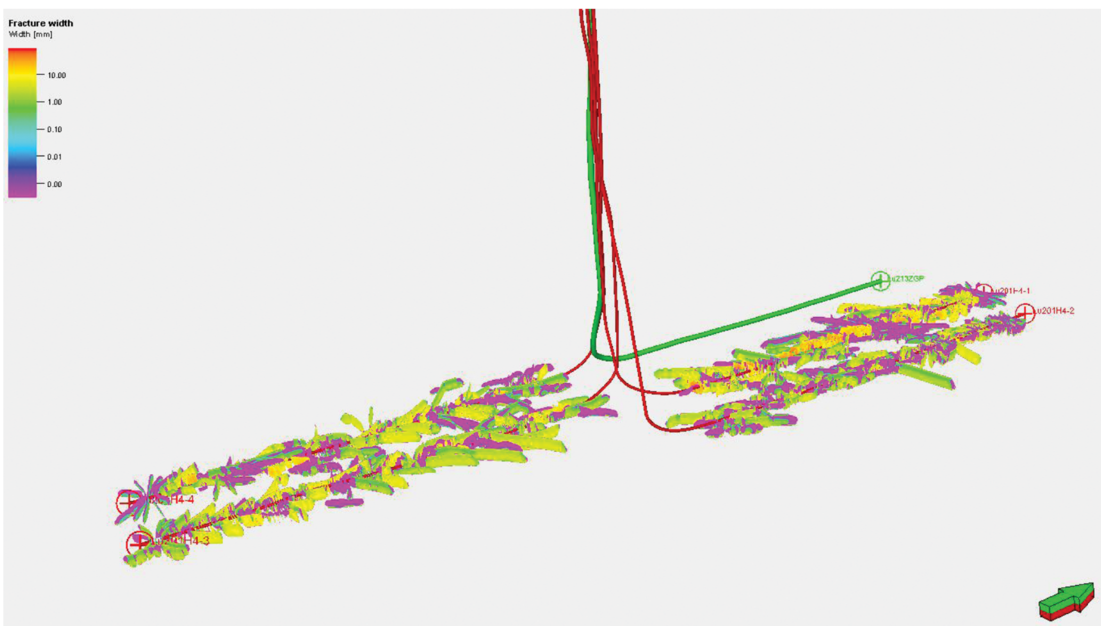

Figure 8—Hydraulic Fracture Simulation Geometry for Well Lu 201H4-4

Conclusions

Based on geological and engineering characteristics of the pad wells and insights from offset completions, a coupled geomechanical model was developed incorporating natural fractures, formation flexure, and fault distributions. Slip-risk evaluations were conducted for six major fracture belts, revealing a platform-wide friction coefficient range of 0.28–0.52. Wells Lu201H4-3 and H4-4 display significant formation flexure, faulting, and localized natural fractures, resulting in high casing-deformation risk. After over 500 days of production, Lu213H has created a pressure-depleted zone, elevating cross-well communication risk for the north cluster (Lu201H4-1 and H4-2). A single fracture belt links Lu213H, H4-1, and H4-2, while two belts traverse the south cluster (H4-3 and H4-4), indicating potential crossflow hazards.

The integrated geoengineering casing deformation prevention system provides essential guidance for lifecycle well planning and field execution, effectively overcoming bottlenecks such as crossflow and casing deformation. This prevention methodology has been deployed across multiple southern Sichuan shale-gas blocks, achieving significant performance gains.

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